Exercise 1: Calculator Program using Method Overloading and Static Variable

Code:

```
class Calculator {

    static int count = 0;

    // Method for integer addition
    int add(int a, int b) {
        count++;
        return a + b;
    }

    // Overloaded method for decimal addition
    double add(double a, double b) {
        count++;
        return a + b;
    }

    // Static method
    static void showCount() {
        System.out.println("Total Calculations : " + count);
    }
}

public class CalculatorDemo {

    public static void main(String[] args) {

        Calculator c = new Calculator();

        System.out.println("Integer Sum : " + c.add(10, 20));
        System.out.println("Decimal Sum : " + c.add(10.5, 20.5));

        Calculator.showCount();
    }
}
```

Output:

```
Integer Sum : 30
Decimal Sum : 31.0
Total Calculations : 2
```

Exercise 2: Restaurant Billing Application using Method Overloading and Static Variable

```
class Restaurant {

    static int totalOrders = 0;

    // Dine-in Bill
    void bill(int amount) {
        totalOrders++;
        System.out.println("Dine-in Bill : " + amount);
    }

    // Takeaway or Delivery Bill
    void bill(int amount, String type) {
        totalOrders++;
        System.out.println(type + " Bill : " + amount);
    }

    // Static method
    static void showOrders() {
        System.out.println("Total Orders : " + totalOrders);
    }
}

public class RestaurantDemo {

    public static void main(String[] args) {

        Restaurant r = new Restaurant();

        r.bill(500);
        r.bill(350, "Takeaway");
        r.bill(450, "Delivery");
```

```
        Restaurant.showOrders();
    }
}
```

Output:

```
Dine-in Bill : 500
Takeaway Bill : 350
Delivery Bill : 450
Total Orders : 3
```